

New US Jet Breaks Sound Barrier in First Flight Test

An experimental aircraft developed by a private American company broke the sound barrier during a recent flight test.

California-based company Boom Supersonic built the plane, named XB-1. The company said the aircraft reached a speed of Mach 1.05 during a test on January 28. The test flight operated from the Mojave Air & Space Port in Mojave, California.

Any speed greater than Mach 1.0 breaks the sound barrier and is considered supersonic. The American space agency NASA explains that at sea level, the speed of sound is estimated to be about 1,236 kilometers per hour.

In a statement, Boom Supersonic said the XB-1 demonstrator aircraft climbed to more than 35,000 feet before reaching Mach 1.0. It recorded the speed within 11 minutes of taking off. The company also released a video feed of the supersonic test flight.

Boom Supersonic XB-1

The company said in a press release, “Historically, supersonic aircraft have been the work of nation states, developed by militaries and governments.” But the latest flight made the XB-1 “the world’s first independently developed supersonic jet” to break the sound barrier, Boom Supersonic said.

The company’s founder and chief executive officer, Blake Scholl, said, “XB-1’s supersonic flight demonstrates that the technology for passenger supersonic flight has arrived.”

Boom Supersonic launched in 2014 “with the goal of making high-speed travel mainstream and enabling a new world of human connection.” With that in mind, the company has been working to develop a supersonic passenger airliner called Overture. The aircraft would carry up to 80 passengers.

Boom Supersonic is one of several American companies seeking to develop supersonic flight vehicles. NASA is also working on a supersonic aircraft in partnership with a U.S. contractor.

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The last time supersonic travel was available to passengers was 2003. That is the year the Concorde – the world’s first supersonic passenger jet aircraft – stopped service. Britain and France jointly operated the Concorde.

But the joint operation between British Airways and Air France suffered several business difficulties that led to the decision to stop flying the supersonic planes. Some of the difficulties were linked to publicity about a deadly Concorde crash in 2000. That incident involved an Air France airliner crashing shortly after taking off from Paris. The crash claimed a total of 113 lives.

Boom Supersonic says its planned Overture airliner will be designed to transport passengers from the U.S. East Coast across the Atlantic to Europe in about 3-and-a-half hours. The flight with normal passenger jets takes about eight hours.

NASA's X-59 supersonic research aircraft sits outside Lockheed Martin's Skunk Works facility in Palmdale, California. (Image Credit: NASA Lockheed Martin Skunk Works)

Amy Marino Spowart is the president and chief executive officer at the National Aeronautic Association. She told The Associated Press the latest supersonic test flight suggests "the future of aviation is here and now." Spowart added, "Not only is there hope for faster and better commercial flight, but Boom proves that it can be done sustainably."

Boom Supersonic has said its planned supersonic airliner will be designed to run on up to 100 percent sustainable aircraft fuel. Such sustainable fuels are made from non-petroleum products, such as food waste or other biomass materials. Both XB-1 and Overture are made mostly from a carbon-based composite material to provide "a strong, lightweight structure," the company said.

NASA's supersonic aircraft – built together with Lockheed Martin – is called X-59. It is also an experimental plane that is seeking to test the level of noise produced by supersonic travel. Currently, supersonic flights are banned over land because of the loud noise they produce.

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NASA has explained it plans to carry out testing this year of the X-59 aircraft, which has been designed and engineered to operate at lower noise levels than other supersonic aircraft. The tests – to take place at supersonic test centers in California – are designed "to prove the quiet supersonic technology works as designed."

NASA said that after that testing is done, the agency plans to begin community reaction studies, beginning in 2026. These will involve flying the X-59 over several cities and then asking people on the ground to share their thoughts on the level of noise they heard.

NASA has said the X-59 – which measures 30 meters long and nine meters wide – is expected to fly at 1.4 times the speed of sound. That is about 1,500 kilometers per hour.

I'm Bryan Lynn.

Quiz - New US Jet Breaks Sound Barrier in First Flight Test

The Associated Press, Reuters and Boom Supersonic reported on this story. Bryan Lynn adapted the reports for VOA Learning English.

Vocabulary

mainstream - adj. กระแสหลัก, ซึ่งเป็นที่ยอมรับของคนส่วนใหญ่

enable - v. ทำให้เป็นไปได้, เปิดโอกาสให้

sustainable - adj. ยั่งยืน, ซึ่งไม่ทำลายสิ่งแวดล้อม

composite - adj. วัสดุผสม, ซึ่งประกอบขึ้นจากวัสดุหลายอย่าง

Reading Comprehension Quiz

1. What significant achievement was made by an experimental aircraft developed by a private American company recently?

- A) It traveled to space.
- B) It broke the sound barrier.
- C) It completed its first commercial flight.
- D) It landed vertically.

2. What is the name of the company that built the XB-1 aircraft?

- A) NASA
- B) Lockheed Martin
- C) Boom Supersonic
- D) Air France

3. What speed did the XB-1 aircraft achieve during its test flight on January 28?

- A) Exactly Mach 1.0
- B) Less than Mach 1.0
- C) Mach 1.05
- D) Over 1,236 kilometers per hour

4. According to NASA, what is the approximate speed of sound at sea level?

- A) Mach 1.0
- B) 35,000 feet per hour
- C) Around 1,236 kilometers per hour
- D) 1.4 times the speed of sound

5. How long after takeoff did the XB-1 record its supersonic speed?

- A) Immediately after climbing to 35,000 feet.
- B) Within 11 minutes.
- C) After several hours of flight.
- D) Before climbing to 35,000 feet.

6. What makes the XB-1's achievement significant in the history of supersonic flight, according to Boom Supersonic?

- A) It was the fastest supersonic jet ever built.
- B) It was the first supersonic jet developed independently by a private company.
- C) It was the first supersonic jet to carry passengers.
- D) It was developed by a military organization.

7. What does Blake Scholl, Boom Supersonic's CEO, believe the XB-1's flight demonstrates?

- A) The end of supersonic travel.
- B) The feasibility of military supersonic jets.
- C) The arrival of technology for passenger supersonic flight.
- D) The need for government funding in aviation.

8. What is Boom Supersonic's ultimate goal with its high-speed aircraft development?

- A) To compete with NASA's X-59 project.
- B) To make high-speed travel mainstream and enable new human connections.
- C) To develop a military reconnaissance aircraft.
- D) To restart the Concorde program.

9. What is the name of the supersonic passenger airliner that Boom Supersonic is currently developing?

- A) XB-1
- B) Concorde
- C) Overture
- D) X-59

10. Who else, besides private American companies like Boom Supersonic, is working on developing supersonic aircraft?

- A) Only European consortia.
- B) The US military exclusively.
- C) NASA, in partnership with a U.S. contractor.

D) British Airways and Air France.

11. When was the last time passengers could travel by supersonic aircraft, and what was the name of that aircraft?

- A) 2014, Overture
- B) 2000, Air France airliner
- C) 2003, Concorde
- D) Currently, XB-1

12. What was a major factor contributing to the Concorde's discontinuation of service?

- A) Lack of passenger interest.
- B) Excessive fuel costs.
- C) Business difficulties and negative publicity from a deadly crash in 2000.
- D) New regulations banning supersonic flight.

13. How much faster is Boom Supersonic's Overture expected to be for transatlantic flights compared to normal passenger jets?

- A) About 8 hours faster.
- B) About 4.5 hours faster.
- C) Only slightly faster, by about an hour.
- D) It will take twice as long.

14. What does Amy Marino Spowart, CEO of the National Aeronautic Association, highlight about Boom's achievement?

- A) It guarantees immediate commercial supersonic flights.
- B) It signifies the future of aviation, especially for faster, better, and sustainable commercial flight.
- C) It demonstrates a breakthrough in military aviation only.
- D) It proves that supersonic flight is too expensive for commercial use.

15. What is a key environmental feature planned for Boom Supersonic's Overture airliner?

- A) It will use conventional jet fuel more efficiently.
- B) It will be designed to run on up to 100 percent sustainable aircraft fuel.
- C) It will primarily use hydrogen as fuel.
- D) It will generate less noise pollution.